13. Limitations of Low-Code Platforms for a Complex Enterprise Use Case

Aim

To analyze the technical, architectural, and operational limitations of low-code development platforms when implemented for complex, large-scale enterprise software requirements, using the Insurance Claim Request & Processing Management System as the application context.

Theory

Low-code platforms such as Zoho Creator enable rapid application development through visual components, predefined templates, and automated workflows. They are suitable for business applications such as insurance claim registration and processing. However, complex enterprise applications may face certain limitations.

1. Customization Constraints

Low-code platforms provide predefined components and workflows. Implementing advanced insurance fraud detection algorithms, complex claim calculations, or highly customized user interfaces may require additional scripting or external services.

2. Scalability and Performance Bottlenecks

As the number of insurance claims and users increases, large volumes of data and concurrent requests may affect application performance. Complex queries and frequent database operations can increase response times.

3. Vendor Lock-in
Applications developed on proprietary platforms may depend on platform-specific forms, workflows, scripts, and databases. Migrating the insurance application to another platform may require considerable redevelopment.

4. Complex Integration Challenges
Integrating the insurance claim application with external insurance databases, payment gateways, legacy systems, or third-party fraud detection services may require custom APIs and additional configuration.

Case Study Scenario: Global Supply Chain & Predictive Logistics Engine

Enterprise Requirement
An insurance company wants to develop a centralized system to manage large numbers of insurance claims, process investigations, calculate settlements, detect fraudulent claims, and integrate with external policy databases.

Platform Chosen
Zoho Creator is selected for rapid development, visual form design, automated workflows, and report generation.

Identified Limitations During Implementation:

1. Customization Wall: Advanced fraud detection algorithms and complex settlement calculations may exceed the capabilities of standard visual workflows and require custom Deluge scripts or external services.

2. Database and Performance Limitations:

Large volumes of claim records and frequent database queries may affect report loading and workflow execution. Query optimization and appropriate data structuring become necessary.

3. Licensing and Scaling Costs:

As the number of users, records, and integrations increases, platform subscription costs and API usage limits may affect the overall cost of maintaining the application.

4. Integration Challenges:

Connecting the system with external policy databases, payment services, and legacy insurance applications may require authenticated connections, API configuration, and additional error handling.

Result

The study demonstrates that low-code platforms such as Zoho Creator are effective for rapidly developing insurance claim management applications. However, customization constraints, performance limitations, vendor dependency, integration complexity, and scaling costs may present challenges as the system grows. Large-scale insurance applications may require external services or traditional software development approaches to achieve greater flexibility, performance, and maintainability.